



POWERED BY AI

Complete Product Training Data

Noupe AI Chatbot Knowledge Base — v1.0

This document is the authoritative reference for the FitAI platform. It covers every page, feature, exercise, workout plan, meal plan, challenge, badge, and user-facing interaction so that an AI assistant can answer any question a FitAI user might ask — accurately and completely.

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1. PRODUCT OVERVIEW & MISSION

What is FitAI?

FitAI is a production-ready, mobile-first web application designed for gym enthusiasts and serious athletes who want intelligent, data-driven fitness guidance. It is built as a comprehensive fitness companion that covers every stage of the training lifecycle — planning, execution, recovery, nutrition, and community accountability.

Core Mission

"Your AI-Powered Fitness Companion" — FitAI exists to stop people from guessing in the gym. It builds personalized workout plans, tracks every rep and PR, and adapts to the user's real life so they keep making progress no matter what.

Target Users

- Gym beginners who need structured, guided programs
- Intermediate lifters who want to track progress and break plateaus
- Advanced athletes who need detailed analytics and progressive overload tracking
- Anyone who wants AI-assisted meal planning aligned to their training goals
- People who benefit from social accountability through community challenges

Key Value Propositions

- Personalized AI workout plans generated in seconds via a step-by-step wizard
- Full exercise library with 100+ curated exercises covering every major muscle group
- Visual progress tracking with charts powered by Recharts
- Gamification: XP points, levels, badges, weekly challenges, and a global leaderboard
- Integrated meal planner with macro breakdowns for 3 goals
- Smart rest timer with AI-suggested rest periods
- Noupe AI chatbot integration point for real-time fitness Q&A

Technology Stack

Frontend:	React 18 + Vite 7
Styling:	Tailwind CSS v4 + shadcn/ui components
Animations:	Framer Motion (page transitions, scroll reveals, micro-interactions)
Routing:	Wouter (lightweight client-side router)
Charts:	Recharts (LineChart for progress tracking)
Forms:	react-hook-form + Zod validation
State:	React useState + localStorage (no backend — prototype mode)
Icons:	Lucide React
Fonts:	Outfit (display) + Inter (body) — Google Fonts

Platform Notes

- FitAI is a prototype: all data is mock/local. No real backend or database exists.
- Authentication is simulated — login/signup stores a mock user in localStorage.
- Progress logs persist in localStorage across browser sessions.
- The app is fully responsive: mobile-first at 375px, tablet and desktop up to 1440px.
- Default theme is dark mode; light mode toggle is available in Profile > Settings.

2. VISUAL DESIGN & THEME

Design Philosophy

FitAI's design is dark, powerful, and energetic — "like a serious gym at midnight." It uses deep blacks and rich dark grays as the base, with electric lime green (#ADFF2F) as the single, vibrant accent color. This communicates performance, precision, and ambition.

Color System

Background (dark):	#0a0a0a — near-black page background
Card (dark):	#111111 — slightly lighter surface for cards
Secondary (dark):	#1a1a1a — muted section backgrounds
Primary Accent:	#ADFF2F / hsl(84 100% 59%) — electric lime green
Primary Text (dark):	#FAFAFA — near-white
Muted Text (dark):	#888888 — secondary descriptions
Border (dark):	#2a2a2a — subtle card borders
Background (light):	#F8F9FA — clean off-white
Primary Accent:	Same lime green in both modes

Typography

Display font:	Outfit — headings, hero text, dashboard titles
Body font:	Inter — body copy, UI labels, descriptions
Monospace:	Built-in monospace — timer countdown display

Layout Patterns

- Fixed top navbar on the landing page with a backdrop-blur glass effect
- Sidebar navigation on desktop (md and above breakpoint)
- Bottom navigation bar on mobile (below md breakpoint)
- Card-based layout for all feature content — consistent padding p-6 or p-7
- Grid system: 1 col mobile, 2 col tablet, 3 col desktop for exercise and feature grids
- Maximum content width of max-w-7xl centered with auto margins

Animation System

- Frame Motion fadeUp variants on all landing page sections (scroll-triggered)
- Page-level transitions: opacity + y-axis slide on route changes
- Staggered children animations on Dashboard cards (0.1s delay each)
- AnimatePresence for wizard step transitions (y-axis slide in/out)
- SVG circular progress stroke animation on the Rest timer
- Pulsing glow effect on the AI chatbot placeholder and timer button

"Powered by AI" Badge

A reusable <PoweredByAI> component appears on: the landing page hero, the dashboard AI insight card, the exercise library smart search, the workout planner header, the meal planner, and the rest timer. It is a small pill badge with a sparkle icon and "POWERED BY AI" text, styled with lime green color on a dark bordered background.

3. APPLICATION ARCHITECTURE

File Structure

<code>src/App.tsx:</code>	Root router, ThemeProvider, Layout wrapper, all route definitions
<code>src/pages/Landing.tsx:</code>	Public landing / marketing page
<code>src/pages/Login.tsx:</code>	Login form with email + password + Google UI
<code>src/pages/Signup.tsx:</code>	Signup form with name, email, password, confirm
<code>src/pages/Dashboard.tsx:</code>	Authenticated home screen with stats and today's workout
<code>src/pages/Exercises.tsx:</code>	Searchable, filterable exercise library
<code>src/pages/ WorkoutPlanner.tsx:</code>	Multi-step AI workout generation wizard
<code>src/pages/Progress.tsx:</code>	Workout logs, charts, and AI performance analysis
<code>src/pages/Challenges.tsx:</code>	Weekly challenges, XP rewards, and leaderboard
<code>src/pages/MealPlanner.tsx:</code>	Goal-based meal suggestions with macros
<code>src/pages/Timer.tsx:</code>	Animated rest timer with presets
<code>src/pages/Profile.tsx:</code>	User profile, settings, theme toggle, achievements
<code>src/components/Layout.tsx:</code>	Shell: sidebar (desktop), bottom nav (mobile), header
<code>src/components/ ThemeProvider.tsx:</code>	Context provider for dark/light mode with localStorage
<code>src/components/ PoweredByAI.tsx:</code>	Reusable "Powered by AI" badge component
<code>src/lib/mockData.ts:</code>	All static mock data: exercises, plans, challenges, meals, badges
<code>src/lib/localStorage.ts:</code>	Helpers for reading/writing progress logs to localStorage

Routing Table

<code>/:</code>	Landing page — public, no auth required
<code>/login:</code>	Login page — public
<code>/signup:</code>	Sign up page — public
<code>/dashboard:</code>	Dashboard — requires mock auth
<code>/exercises:</code>	Exercise library — accessible to all
<code>/workout-planner:</code>	Workout plan generator wizard
<code>/progress:</code>	Progress tracker with charts
<code>/challenges:</code>	Community challenges + leaderboard
<code>/meal-planner:</code>	Nutrition planner
<code>/timer:</code>	Rest timer
<code>/profile:</code>	Profile and settings

Data Storage

FitAI uses no backend database. All persistent state lives in the browser's localStorage. The `localStorage.ts` library provides three main functions:

- `getProgressLogs()` — returns array of logged workout entries
- `addProgressLog(entry)` — appends a new log entry and persists to localStorage
- `getMockUser() / setMockUser()` — reads and writes the mock authenticated user

Mock data for exercises, workout plans, challenges, meal plans, and badges lives in `mockData.ts` and is imported directly into components. It never changes unless the developer edits the file.

4. PAGES & USER FLOWS

4.1 Landing Page (/)

The public-facing marketing page. Fully scrollable with 9 distinct sections:

- Navbar — fixed glass bar with 'Log in' and 'Sign up' free CTA buttons
- Hero — headline "Your AI-Powered Fitness Companion", subtext, two CTA buttons (Start for Free !' / sign up, Explore Exercises !' / exercises), 5 avatar avatars with "Join 50,000+ athletes" trust line
- Stats Bar — 50K+ Active Athletes, 2M+ Workouts Logged, 98% Goal Achievement, 4.9 App Rating
- Features Grid — 6 feature cards (AI Workout Plans, Progress Analytics, Adaptive Training, Meal Planner, Challenges & XP, Smart Rest Timer)
- How It Works — 4 numbered steps with a connecting line: Set Your Goal !' Get Your Plan !' Train & Track !' See Results
- Progress Callout — 4 sample stat cards (Bench Press +22 lbs, Total Volume 142K lbs, Workout Streak 34 days, Goal Progress 78%)
- Challenges Callout — 3 sample challenge cards with XP rewards and participant counts
- Testimonials — 3 user reviews with name, role, 5-star ratings and quotes
- Footer — 4 columns of 8 platform features with a CTA
- Noupe AI Chatbot placeholder — styled card with dashed border and <div id="noupe-ai-chatbot">
- Final CTA — "Ready to train smarter?" section with two buttons and footer

4.2 Login Page (/login)

- Email input field (type=email, required)
- Password input field (type=password, required)
- "Continue with Google" button — UI only, no real OAuth
- "Log in" submit button — on success stores mock user in localStorage and redirects to /dashboard
- Link to /signup for new users
- Form validated with react-hook-form + Zod

4.3 Sign Up Page (/signup)

- Full name input
- Email input
- Password input
- Confirm password input (must match password)
- "Continue with Google" button — UI only
- "Create Account" submit button — redirects to /dashboard
- Link to /login for returning users

4.4 Dashboard (/dashboard)

The authenticated home screen — the first thing a logged-in user sees.

- Header: "Welcome back, [FirstName]" greeting, streak counter (14 days), level + XP display (Level 12, 4500 XP)
- AI Insight card: highlighted card with brain icon and a personalized training tip
- Today's Plan card: Push Day — Chest & Triceps with 4 exercises and a Start button
- Quick access grid: 4 shortcut cards linking to Library, Progress, Planner, Challenges
- Level Progress bar: shows progress to next level (65% complete, 1,500 XP needed)
- Noupe AI Chatbot panel: right column card with <div id="noupe-ai-chatbot"> inside

4.5 Exercise Library (/exercises)

- Page heading and subtitle
- Smart Search card (AI-badged) — text input filtering by name and description
- Sample placeholder: show chest exercises with dumbbells for beginners
- Clear Filters button — appears only when any filter is active
- Responsive grid (1, 2, 3 columns) of exercise cards
- Each card contains: YouTube demo video embed, exercise name, difficulty badge, description, muscle group, tag, equipment tag
- Empty state: "No exercises found matching your criteria."

4.6 Workout Planner (/workout-planner)

- Step 0 — Primary Goal: Muscle Gain | Fat Loss | Strength | Endurance (radio buttons)
- Step 1 — Experience Level: Beginner | Intermediate | Advanced
- Step 2 — Equipment: Full Gym | Dumbbells Only | Bodyweight
- Step 3 — "Synthesizing Your Plan" loading animation (2 second delay)
- Result card: plan name, description, 4 star tiles (goal/level/days/equipment), AI Notes paragraph, two buttons: "Start This Plan" and "Copy Summary"
- Progress bar at top of wizard card animates from 0% to 100% across steps

4.7 Progress Tracker (/progress)

- "Log Workout" button opens a modal dialog with fields: exercise name, sets, reps, weight (lbs)
- Bench Press 1.5K collection chart: line graph with weekly data (200 lbs over 5 weeks)
- Recent History sidebar: ever-sealed chronological log of exercises with date, sets × reps, weight
- All logs persist in localStorage so they survive page refresh

4.8 Community Challenges (/challenges)

- 5 active weekly challenges: 100 Pushups a Day (500 XP), Run 5k (300 XP), 7-Day Plank Challenge (400 XP), Squat Your Bodyweight (600 XP), No Sugar Week (200 XP)
- Each challenge shows: type badge, participant count, title, XP reward, and a "Complete Challenge" button
- Completing a challenge triggers a toast notification with XP earned and changes the button to

"Completed"

Global Leaderboard sidebar: top 5 users with XP — current user (Alex Fitness) is highlighted at

- rank 3
- Streak encouragement card: "You're in the top 15% of users this week."
-

4.9 Meal Planner (/meal-planner)

Goal selection with 3 radio buttons: Muscle Gain, Fat Loss, Maintenance

Macro overview icons: Protein, Carbs, Fats

- Daily meal suggestions update dynamically based on selected goal
- Each meal entry shows: meal name (Breakfast/Lunch/Dinner), food suggestion, macro breakdown (P/C/F grams)
- Hydration reminder footer card: "Aim for at least 3-4 liters of water per day"
-

4.10 Rest Timer (/timer)

AI suggestion pill: "AI Suggests: 90s for hypertrophy goals"

Large SVG circular timer 256x256px with lime green progress stroke

MVFS countdown display in monospace font

- Controls: circular Reset button, large circular Play/Pause button with animated icon swap
- 5 preset buttons: 30s, 1m, 1m30s, 2m, 3m
- On completion: toast notification "Rest complete! Time for your next set." + device vibration (if supported)
-

4.11 Profile & Settings (/profile)

User avatar: circular, initials (AF for Alex Fitness), Level 12 Elite badge

Two tabs: "Settings & Profile" and "Achievements"

- Settings tab: Edit Profile form (name + email), App Preferences card (dark mode toggle, push notifications toggle)
- Danger Zone: "Delete Account" button (UI only)
- Achievements tab: 5 earned badges (First Workout, 7-Day Streak, 1000 XP Club, Strength Master, Early Bird) + 1 locked badge (Iron Master)
-

5. EXERCISE LIBRARY — COMPLETE CATALOGUE

FitAI includes 10 curated exercises covering all major muscle groups. Each exercise has a video demo, description, muscle group tag, equipment tag, and difficulty rating.

E1: Barbell Bench Press

Muscle Group: Chest
Equipment: Barbell
Difficulty: Intermediate

A compound pushing exercise targeting the pectoralis major, anterior deltoids, and triceps. Performed lying on a flat bench, gripping the barbell at shoulder-width, lowering to the chest and pressing back up. Considered the gold-standard upper body strength exercise.

E2: Dumbbell Flyes

Muscle Group: Chest
Equipment: Dumbbell
Difficulty: Beginner

An isolation exercise targeting the pectoral muscles. Performed lying on a flat bench, holding dumbbells above the chest with slightly bent elbows, lowering them in a wide arc and squeezing back together. Excellent for chest development and stretching the pecs under load.

E3: Pull-ups

Muscle Group: Back
Equipment: Bodyweight
Difficulty: Advanced

A challenging upper-body compound exercise targeting the latissimus dorsi and biceps. Hanging from a bar with an overhand grip, pulling the body up until the chin passes the bar. One of the best indicators of relative upper-body strength.

E4: Barbell Squat

Muscle Group: Legs
Equipment: Barbell
Difficulty: Intermediate

The king of lower-body compound exercises. The barbell rests across the upper traps or rear delts. The athlete squats to at least parallel depth, targeting quadriceps, hamstrings, glutes, and erector spinae. Builds overall lower-body mass and strength.

E5: Leg Press

Muscle Group: Legs
Equipment: Machine
Difficulty: Beginner

A machine-based lower-body exercise where the user pushes a weighted platform away using both legs. Safer for beginners due to spinal support. Targets quads primarily, with secondary activation of hamstrings and glutes depending on foot placement.

E6: Overhead Press

Muscle Group:	Shoulders
Equipment:	Barbell
Difficulty:	Intermediate

A standing compound press where the barbell is pressed vertically from shoulder height to lockout overhead. Primarily targets the anterior and medial deltoids, with significant triceps and upper trap involvement. Also builds core stability.

E7: Dumbbell Lateral Raise

Muscle Group:	Shoulders
Equipment:	Dumbbell
Difficulty:	Beginner

An isolation exercise for the medial (side) deltoid head. Performed standing, raising dumbbells laterally to shoulder height with slightly bent elbows. Creates the "capped shoulders" aesthetic. Best performed with controlled tempo and moderate weight.

E8: Bicep Curl

Muscle Group:	Arms
Equipment:	Dumbbell
Difficulty:	Beginner

The classic arm isolation exercise. Holding dumbbells at the sides, curling both or alternating arms up toward the shoulder while keeping elbows fixed. Targets the biceps brachii and brachialis. Can be performed seated or standing.

E9: Tricep Pushdown

Muscle Group:	Arms
Equipment:	Machine
Difficulty:	Beginner

A cable machine isolation exercise for the triceps. Gripping a straight or V-bar attachment, the user pushes it downward while keeping elbows pinned to the sides. Excellent for tricep hypertrophy and finishing a push-day session.

E10: Plank

Muscle Group:	Core
Equipment:	Bodyweight
Difficulty:	Beginner

A foundational static core stabilization exercise. Holding a straight-body position face-down, supported on forearms and toes. Builds deep core stability, transverse abdominis strength, and spinal endurance. Duration-based: typically 30–60+ seconds.

Filter Options Available

By Muscle Group:	All, Chest, Back, Legs, Shoulders, Arms, Core
By Equipment:	All, Barbell, Dumbbell, Bodyweight, Machine
By Difficulty:	All, Beginner, Intermediate, Advanced

Users can also use the Smart Search bar to search by keyword (exercise name or description).

6. WORKOUT PLANS — COMPLETE CATALOGUE

The AI Workout Generator creates one of 3 sample plans based on the user's goal, level, equipment, and days per week.

WP1: Push/Pull/Legs Hypertrophy

Primary Goal: Muscle Gain
Level: Intermediate
Days Per Week: 6
When Generated: Selected when the user's goal is "muscle gain".

A 6-day PPL split optimized for maximum muscle hypertrophy. Each week the trainee performs two Push days (chest, shoulders, triceps), two Pull days (back, biceps), and two Leg days. This high-frequency approach ensures each muscle group is trained twice per week for optimal protein synthesis stimulus.

Sample Structure:

Day 1 (Push): Bench Press, Incline DB Press, OHP, Lateral Raise, Tricep Pushdown
Day 2 (Pull): Barbell Row, Pull-ups, Face Pull, Bicep Curl, Hammer Curl
Day 3 (Legs): Barbell Squat, Romanian Deadlift, Leg Press, Leg Curl, Calf Raise
Day 4-6: Repeat with variation in exercise selection

WP2: Full Body Strength

Primary Goal: Strength
Level: Beginner
Days Per Week: 3
When Generated: Selected when goal is "strength" OR when user is a beginner.

A 3-day full-body program built around the major compound lifts. Designed for beginners and anyone returning from a break. Each session trains the entire body with emphasis on progressive overload across the squat, hinge, push, and pull movement patterns. Rest days between each session allow full recovery.

Sample Structure:

Day 1 (Mon): Barbell Squat 3x5, Bench Press 3x5, Barbell Row 3x5
Day 2 (Wed): Deadlift 1x5, Overhead Press 3x5, Pull-ups 3xmax
Day 3 (Fri): Barbell Squat 3x5, Bench Press 3x5, Barbell Row 3x5

WP3: HIIT Fat Burner

Primary Goal: Fat Loss
Level: Intermediate
Days Per Week: 4
When Generated: Selected when the user's goal is "fat loss".

A 4-day high-intensity interval training program designed to maximize caloric expenditure and metabolic conditioning. Combines resistance training with cardio intervals to create an "afterburn" (EPOC) effect that continues burning calories for hours post-workout.

Sample Structure:

Day 1: Lower body HIIT circuits — squat jumps, lunges, box steps, sled push
Day 2: Upper body conditioning — push-ups, rows, burpees, battle ropes
Day 3: Rest or active recovery
Day 4: Full body HIIT — deadlifts, jump rope, KB swings, plank variations
Day 5: Cardio intervals — 20 min HIIT on bike or treadmill

How Plan Selection Works

The wizard matches the user's stated goal to the closest plan. If goal = "muscle gain" ! WP1. If goal = "strength" ! WP2. If goal = "fat loss" ! WP3. If goal = "endurance", the fallback is WP1. Future versions will

generate truly dynamic plans; in the prototype, these 3 plans serve as templates.

7. MEAL PLANS & NUTRITION

FitAI's Meal Planner provides daily meal suggestions tailored to three primary goals. Each suggestion includes a macro breakdown showing approximate protein (P), carbohydrates (C), and fats (F) in grams.

Goal: Muscle Gain (Caloric Surplus)

Breakfast

Oatmeal with protein powder, berries, and peanut butter

Macros: P: 35g, C: 60g, F: 15g

High-carb, high-protein breakfast to fuel morning workouts and kick-start muscle protein synthesis.

Lunch

Chicken breast, sweet potato, and broccoli

Macros: P: 45g, C: 50g, F: 5g

Lean protein with complex carbs and micronutrient-rich vegetables. Classic muscle-building plate.

Dinner

Salmon, quinoa, and asparagus

Macros: P: 40g, C: 45g, F: 18g

Omega-3 rich protein supports joint health and inflammation reduction post-training. Quinoa provides complete amino acids.

Goal: Fat Loss (Caloric Deficit)

Breakfast

Egg white scramble with spinach and tomatoes

Macros: P: 25g, C: 5g, F: 2g

Very low calorie, high protein breakfast that preserves lean muscle while in a deficit.

Lunch

Grilled turkey salad with vinaigrette

Macros: P: 35g, C: 15g, F: 10g

Lean protein with fiber-rich vegetables and healthy fats from olive oil. Keeps hunger at bay.

Dinner

White fish with cauliflower rice

Macros: P: 30g, C: 20g, F: 8g

White fish (cod or tilapia) is extremely low in calories and high in protein. Cauliflower rice replaces carb-heavy white rice.

Goal: Maintenance (Body Recomposition)

Breakfast

Greek yogurt with almonds and honey

Macros: P: 20g, C: 30g, F: 12g

Balanced, convenient breakfast. Greek yogurt provides casein protein for sustained release.

Lunch

Beef wrap with mixed greens

Macros: P: 30g, C: 40g, F: 15g

Moderate protein, balanced macros. Good pre-workout lunch option.

Dinner

Chicken stir-fry with mixed vegetables

Macros: P: 35g, C: 35g, F: 10g

High-protein, versatile dinner with micronutrient-dense vegetables. Works for any training goal.

Hydration Guidance

FitAI recommends athletes drink at least 3–4 liters of water per day, with increased intake around the workout window. Staying hydrated is critical for performance, recovery, and nutrient transport.

Pre- and Post-Workout Nutrition (General Principles)

Pre-Workout (60–90 minutes before):

- Muscle Gain: complex carbs + moderate protein (e.g., oats + protein shake)
- Fat Loss: light protein + minimal carbs (e.g., egg whites + small fruit)
- Maintenance: balanced snack (e.g., banana + peanut butter)

Post-Workout (within 30–60 minutes):

- Protein is the priority to initiate muscle protein synthesis
- Muscle Gain: fast-acting carbs + whey protein shake
- Fat Loss: lean protein only (e.g., chicken breast, tuna)
- Maintenance: protein + moderate carbs to replenish glycogen

8. COMMUNITY CHALLENGES & XP SYSTEM

FitAI's Community Challenges feature drives accountability and competitive motivation through weekly fitness challenges, XP rewards, and a global leaderboard. Challenges are described as "AI-curated" — in the prototype, 5 static challenges are always active.

Active Weekly Challenges (Prototype Data)

C1: 100 Pushups a Day

Challenge Type: Strength

XP Reward: +500 XP

Current Participants: 1,240

Complete 100 push-ups in a single day. Can be split across multiple sets throughout the day. Targets chest, triceps, and anterior delts.

C2: Run 5k

Challenge Type: Endurance

XP Reward: +300 XP

Current Participants: 890

Complete a 5-kilometer run at any pace, outdoors or on a treadmill. Builds cardiovascular base and mental toughness.

C3: 7-Day Plank Challenge

Challenge Type: Core

XP Reward: +400 XP

Current Participants: 2,100

Complete a plank hold every day for 7 consecutive days. Duration goals progress from 30 seconds on day 1 to 90+ seconds by day 7.

C4: Squat Your Bodyweight

Challenge Type: Strength

XP Reward: +600 XP

Current Participants: 450

Perform a barbell back squat with a load equal to your own bodyweight for at least one rep. Celebrates the milestone of squatting bodyweight, a key strength benchmark.

C5: No Sugar Week

Challenge Type: Nutrition

XP Reward: +200 XP

Current Participants: 3,200

Go 7 consecutive days without added sugar. Promotes dietary discipline and blood sugar regulation. Particularly useful during fat loss phases.

Global Leaderboard (Sample Data)

#1 — Sarah J.:	12,400 XP
#2 — Mike T.:	11,200 XP
#3 — Alex Fitness (current user):	4,500 XP
#4 — Chris D.:	4,200 XP
#5 — Emma W.:	3,800 XP

The leaderboard resets weekly. Users earn XP by completing challenges. The current user (Alex Fitness) is displayed in the leaderboard with a highlighted row and is described as being in the "top 15% of users this week."

XP Earning Summary

- 100 Pushups a Day: +500 XP
- Run 5k: +300 XP
- 7-Day Plank Challenge: +400 XP
- Squat Your Bodyweight: +600 XP
- No Sugar Week: +200 XP
- Maximum weekly XP from challenges: 2,000 XP

9. PROGRESS TRACKING SYSTEM

The Progress Tracker is FitAI's analytics hub. Users log workouts manually and view their progress over time via charts. AI-generated analysis text provides actionable coaching insights.

Progress Log Entry Fields

Exercise Name:	Free-text string (e.g., "Barbell Bench Press")
Date:	Auto-populated as today's date (YYYY-MM-DD)
Sets:	Integer number of sets performed
Reps:	Integer number of reps per set
Weight:	Weight in pounds (use 0 for bodyweight exercises)

Default Progress Log Entries (Mock Data)

2023-10-01: Barbell Bench Press — 3 sets × 8 reps @ 185 lbs
2023-10-03: Barbell Squat — 3 sets × 6 reps @ 225 lbs
2023-10-05: Pull-ups — 3 sets × 10 reps @ Bodyweight
2023-10-08: Barbell Bench Press — 3 sets × 8 reps @ 190 lbs
2023-10-10: Barbell Squat — 3 sets × 6 reps @ 235 lbs

Bench Press Chart Data (1RM Projection)

The Progress page displays a line chart showing estimated bench press 1RM over 5 weeks:

Week 1: 185 lbs
Week 2: 190 lbs
Week 3: 190 lbs
Week 4: 195 lbs
Week 5: 200 lbs

AI Performance Analysis Messages

The following AI analysis text appears on the Progress page:

"Your bench press has plateaued at 190lbs for two weeks. This indicates central nervous system fatigue. Recommendation: Take a deload week on push movements, dropping volume by 40%."

On the Dashboard, the AI Insight card reads: "Based on your last 3 workouts, your bench press form might be breaking down on the 4th set. Try dropping the weight by 5% and focus on a slow eccentric motion today."

These messages are static in the prototype but represent the type of AI coaching FitAI will provide when a real AI engine is integrated.

Body Measurements (Planned)

The Progress page includes a body measurements section (UI placeholder in the prototype). Future versions will track: weight, body fat %, chest, waist, hips, arms, and legs measurements over time.

10. GAMIFICATION: XP, LEVELS & BADGES

XP System

FitAI uses an experience points (XP) system to reward consistent training and challenge completion. XP accumulates over time and determines the user's level. The mock user (Alex Fitness) starts at Level 12 with 4,500 XP.

Current User XP:	4,500
Current Level:	12 (Elite)
XP to Next Level:	1,500 XP needed for Level 13
Progress:	65% of the way to Level 13

Leveling System

Each level requires incrementally more XP than the last. Level progression is shown as a progress bar on both the Dashboard and Profile pages. Level title at Level 12 is "Elite" — indicating high experience. Higher levels unlock new badges and challenge types (future feature).

Badge System — Complete Catalogue

Users earn badges as milestones and achievements. All 5 earned badges are displayed in the Profile > Achievements tab.

First Workout

Awarded for logging your very first workout in FitAI. Entry-level achievement to encourage new users.

7-Day Streak

Awarded for logging workouts or visiting the app for 7 consecutive days. Encourages daily habit formation.

1000 XP Club

Awarded when the user accumulates 1,000 XP total. Marks the transition from casual to committed user.

Strength Master

Awarded for demonstrating consistent progress in compound strength exercises (e.g., squatting bodyweight or benching 225 lbs).

Early Bird

Awarded for completing 5 workouts before 8:00 AM. Celebrates dedication and morning training consistency.

Iron Master (LOCKED)

A locked badge not yet earned by the user. Requirements are not shown in the UI — building anticipation.

Streak System

FitAI tracks workout streaks — consecutive days of training activity. The mock user has a 14-day streak displayed prominently on the Dashboard header. A streak counter with a flame icon creates urgency to maintain the streak. Future: streak breaks reset the counter; streak shields can protect it.

11. REST TIMER

The Rest Timer at /timer gives athletes a precision tool for managing recovery between sets. AI suggests an optimal rest duration based on the user's training goal.

AI Suggestion

"AI Suggests: 90s for hypertrophy goals" — this message appears in a lime green pill badge above the timer. Rest period guidance is based on the training goal set in the user's profile.

Rest Duration Science (behind the AI suggestions)

- 30 seconds — appropriate for light isolation work, supersets targeting different muscles, or endurance circuits
- 60 seconds — standard for moderate hypertrophy work (10–15 rep range)
- 90 seconds — optimal for hypertrophy (8–12 rep range); allows partial ATP-PC system recovery
- 2 minutes — heavy hypertrophy / power sets (6–8 reps)
- 3 minutes — strength-focused sets (1–5 reps); allows near-complete CNS and phosphocreatine recovery

Timer Controls

Preset Buttons:	30s, 1m, 1m30s (default), 2m, 3m
Play/Pause:	Large circular button — toggles the countdown
Reset:	Smaller circular button — resets to the current preset duration
Completion:	Toast notification: "Rest complete! Time for your next set." + device vibration on mobile
Visual:	SVG circle with lime green arc that depletes as time counts down

12. PROFILE & SETTINGS

Mock User Profile Data

Name:	Alex Fitness
Email:	alex@example.com
Level:	12
XP:	4,500
Streak:	14 days
Primary Goal:	Muscle Gain

Editable Profile Fields

- Full Name (text input)
- Email address (email input)
- Both fields are pre-filled from mockData and can be updated
- "Save Changes" triggers a success toast: "Profile updated successfully"

App Preferences

Dark Mode Toggle:	Switch — toggles dark/light theme. Syncs to localStorage. Default: dark.
Push Notifications:	Switch — UI only, no real notifications. Default: enabled.

Danger Zone

"Delete Account" button is present but performs no action in the prototype. It is styled in the destructive (red) color scheme with a warning message about permanent data loss.

Theme Behavior

- ThemeProvider wraps the entire app and manages a "theme" state value ("dark" or "light")
- Theme is persisted in localStorage under the key "fita-theme"
- On mount, the ThemeProvider reads from localStorage and applies the class "dark" to document.documentElement
- The dark/light toggle on the Profile page calls setTheme() from the useTheme() hook

13. AI FEATURES & NOUPE AI INTEGRATION

FitAI's AI Philosophy

FitAI is described as "AI-powered" throughout the product. In the prototype, AI features are represented by static text outputs and smart UI patterns that simulate AI behavior. The platform is architecturally ready for a real AI engine to replace these static values. The Noupe AI chatbot is the primary real-AI integration point.

Current AI-Simulated Features

- AI Insight on Dashboard: "Based on your last 3 workouts, your bench press form might be breaking down on the 4th set."
- AI Performance Analysis on Progress: plateau detection + deload recommendation
- AI Workout Generator: the wizard generates a "personalized" plan from static templates based on user inputs
- AI Notes on generated plan: paragraph explaining why the specific plan was chosen for the user
- AI Suggestion on Rest Timer: AI Suggests 90s for hypertrophy goals
- AI Suggestion on Exercise Library: labeled as AI-powered (client-side keyword filtering)
- "Powered by AI" badge: appears on 6+ locations throughout the app
- Adjustment suggestion on Progress: "increase bench press by 5% next week"

Noupe AI Chatbot Integration Point

FitAI contains two dedicated mounting points for the Noupe AI chatbot. Both are implemented as div elements with the id "noupe-ai-chatbot" — the standard mounting convention for embedding external chatbot widgets.

Location 1: Landing Page (/), Noupe AI Section

A full-width section with the heading "Ask anything. Get expert answers instantly." and the subtext "Form checks, substitute exercises, nutrition questions — your AI coach is always on call." The chatbot div is styled with a dashed border, min-height of 260px, and centered placeholder content.

Location 2: Dashboard (/dashboard), Right Column

A card in the right sidebar column titled "FitAI Assistant" with a "Powered by AI" badge. The chatbot div fills the card with min-height ~400px. This is the in-app persistent access point so users can ask questions without leaving the dashboard.

Embedding the Noupe Chatbot

To embed the Noupe AI chatbot into FitAI:

- Locate the two `<div id="noupe-ai-chatbot">` elements in Landing.tsx and Dashboard.tsx
- Follow Noupe's embedding instructions (typically a script tag + initialization call)
- The existing styling (border, rounded corners, background) wraps the chatbot for visual consistency
- No other code changes are needed — the integration points are purpose-built and production-ready

Questions Noupe AI Should Be Able to Answer About FitAI

- How do I generate a workout plan? ! Go to /workout-planner and complete the 4-step wizard
- How do I log a workout? ! Visit /progress and click "Log Workout"
- What exercises does FitAI have? ! 10 exercises across chest, back, legs, shoulders, arms, and core
- How do I earn XP? ! Complete community challenges at /challenges
- What is my current level? ! Level 12, shown on the dashboard
- How do I change to light mode? ! Visit /profile > Settings > dark mode toggle
- What meal plan should I follow? ! Visit /meal-planner and select your goal
- How long should I rest between sets? ! Use the timer at /timer (AI suggests 90s for hypertrophy)
- What badges can I earn? ! First Workout, 7-Day Streak, 1000 XP Club, Strength Master, Early Bird

14. FAQ — QUESTIONS USERS WILL ASK

This section is the primary training data for the Noupe AI chatbot. It covers every question a FitAI user is likely to ask, with accurate, detailed answers grounded in the actual product.

General App Questions

Q: What is FitAI?

A: FitAI is an AI-powered fitness web app that helps you discover exercises, generate personalized workout plans, track your progress, follow community challenges, and get nutrition guidance — all in one place.

Q: Is FitAI free to use?

A: Yes, FitAI is completely free during the prototype phase. All features are available with no credit card required.

Q: Do I need to create an account to use FitAI?

A: You can browse the exercise library without an account. For the dashboard, progress tracking, workout planner, and challenges, you'll need to sign up.

Q: Does FitAI work on my phone?

A: Yes. FitAI is mobile-first and works on any device — phone, tablet, or desktop. On mobile you'll see a bottom navigation bar instead of a sidebar.

Q: Is my data saved?

A: Your progress logs and settings are saved locally in your browser. This means they persist across visits but won't carry over if you use a different browser or device.

Exercise Library Questions

Q: How many exercises are in the library?

A: FitAI currently has 10 exercises covering all major muscle groups: chest, back, legs, shoulders, arms, and core.

Q: Can I search for exercises?

A: Yes. Use the Smart Search bar at the top of the Exercise Library page. You can search by name, description, or natural language like "chest exercises with dumbbells for beginners".

Q: How do I filter exercises by muscle group?

A: Use the Muscle Group dropdown filter. Options include: All, Chest, Back, Legs, Shoulders, Arms, and Core.

Q: What equipment options are available?

A: You can filter exercises by: Barbell, Dumbbell, Bodyweight, and Machine.

Q: What does each exercise card show?

A: Each card shows a video demo, the exercise name, difficulty badge (Beginner/Intermediate/Advanced), a brief description, muscle group tag, and equipment tag.

Workout Plan Questions

Q: How does the Workout Plan Generator work?

A: Go to /workout-planner and answer 4 questions: your goal, experience level, available equipment, and how many days per week you can train. FitAI then generates a personalized workout plan.

Q: What workout plans are available?

A: 3 plans: Push/Pull/Legs Hypertrophy (muscle gain, 6 days/week), Full Body Strength (strength/beginner, 3 days/week), and HIIT Fat Burner (fat loss, 4 days/week).

Q: Which plan should I pick for muscle gain?

A: FitAI will recommend the Push/Pull/Legs Hypertrophy plan — a 6-day split that trains each muscle group twice per week for maximum growth stimulus.

Q: I'm a beginner — what plan is best for me?

A: FitAI will recommend the Full Body Strength plan — a 3-day program focused on compound lifts with full recovery between sessions. Perfect for building a strong foundation.

Q: Can I share my workout plan?

A: Yes. After the plan is generated, click "Copy Summary" to copy the plan details to your clipboard, which you can then share anywhere.

Progress Tracking Questions

Q: How do I log a workout?

A: Go to /progress and click the "Log Workout" button. Fill in the exercise name, number of sets, reps, and weight, then click Save.

Q: Are my workout logs saved permanently?

A: They are saved in your browser's localStorage, so they persist across page refreshes. They won't sync across devices or browsers.

Q: What does the progress chart show?

A: The chart shows your estimated Bench Press 1RM (one-rep max) progression over 5 weeks, giving you a visual view of your strength curve.

Q: What does the AI suggest on the Progress page?

A: FitAI analyzes your recent logs and provides coaching insights, such as detecting plateaus and recommending a deload week when progress stalls.

Q: What are bodyweight exercises logged as?

A: Enter 0 in the weight field for bodyweight exercises. FitAI displays these as "BW" in the log.

Challenges & XP Questions

Q: How do I complete a challenge?

A: Go to /challenges, find a challenge you want to complete, and click the "Complete Challenge" button. You'll receive an XP reward and a confirmation toast.

Q: How much XP do challenges reward?

A: 100 Pushups a Day: 500 XP, Run 5k: 300 XP, 7-Day Plank Challenge: 400 XP, Squat Your Bodyweight: 600 XP, No Sugar Week: 200 XP.

Q: Where is the leaderboard?

A: On the /challenges page in the right sidebar. It shows the top 5 users by XP for the current week. Your position is highlighted.

Q: What is my current ranking?

A: The mock user (Alex Fitness) is ranked #3 on the global leaderboard with 4,500 XP.

Q: How do I level up?

A: Earn XP by completing challenges. The dashboard shows your progress bar to the next level. The current user needs 1,500 more XP to reach Level 13.

Meal Planner Questions

Q: How does the Meal Planner work?

A: Go to /meal-planner and select your primary goal (Muscle Gain, Fat Loss, or Maintenance). FitAI instantly shows a daily meal plan with breakfast, lunch, and dinner suggestions and macro breakdowns.

Q: What should I eat for muscle gain?

A: For muscle gain: Breakfast — oatmeal with protein powder, berries, and peanut butter (P:35g C:60g F:15g). Lunch — chicken breast, sweet potato, broccoli (P:45g C:50g F:5g). Dinner — salmon, quinoa, asparagus (P:40g C:45g F:18g).

Q: What should I eat for fat loss?

A: For fat loss: Breakfast — egg white scramble with spinach and tomatoes (P:25g C:5g F:2g). Lunch — grilled turkey salad with vinaigrette (P:35g C:15g F:10g). Dinner — white fish with cauliflower rice (P:30g C:20g F:8g).

Q: How much water should I drink?

A: FitAI recommends 3–4 liters of water per day, especially around your workout window.

Rest Timer Questions

Q: How do I use the Rest Timer?

A: Go to /timer. Select a preset duration (30s, 1m, 1m30s, 2m, or 3m), then press the Play button. It counts down and notifies you when rest is over.

Q: How long should I rest between sets?

A: FitAI's AI suggests 90 seconds for hypertrophy goals. For strength training (heavy compound lifts), rest 2–3 minutes. For endurance circuits, 30–60 seconds is appropriate.

Q: Does the timer make a sound when it ends?

A: Yes — a toast notification pops up saying "Rest complete! Time for your next set." On mobile devices that support it, your phone will also vibrate.

Profile & Account Questions

Q: How do I switch to light mode?

A: Go to /profile, select the "Settings & Profile" tab, and toggle the "Dark Mode" switch to off. The change applies instantly and is remembered.

Q: How do I update my profile information?

A: Go to /profile > Settings & Profile tab. Edit your name or email in the "Edit Profile" form and click "Save Changes".

Q: What badges have I earned?

A: The current mock user has earned: First Workout, 7-Day Streak, 1000 XP Club, Strength Master, and Early Bird. One badge (Iron Master) is still locked.

Q: What is my current streak?

A: The mock user has a 14-day streak, shown on the Dashboard with an orange flame icon.

Q: How do I delete my account?

A: Go to /profile > Settings & Profile > Danger Zone and click "Delete Account". Note: in the prototype, this button does not perform any real action.

Technical & Troubleshooting Questions

Q: Why did my logged workouts disappear?

A: Progress logs are stored in your browser's localStorage. They can be cleared if you clear your browser data, use private/incognito mode, or switch to a different browser.

Q: Can I use FitAI offline?

A: The app loads from the web, so an initial internet connection is required. Once loaded, most features (timer, exercise library, progress logs) work without an active connection since data is stored locally.

Q: Why is the AI chatbot showing as a placeholder?

A: The Noupe AI chatbot is an external integration that will be embedded once fully configured. The placeholder marks exactly where it will appear. The app is ready for the integration.

Q: How do I navigate between pages?

A: On desktop: use the sidebar on the left. On mobile: use the bottom navigation bar. You can also navigate directly to any page by entering its URL path.

FitAI Training Data — End of Document

This document covers 100% of FitAI's features, pages, data, and user interactions.

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